How to Debug a Slow API: Interview Guide

🔥 About ME

Hey, I'm Shubham, 7+ years in tech, 100+ interviews on both sides of the table.

Career path: Infosys → Pubmatic → Amazon → Coursera (currently WFH)

I've cracked multiple high-paying offers across India and internationally using the exact playbook below.

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Introduction

When an interviewer asks *You have a slow API what do you do?*, they're testing your **debugging mindset** and **real-world experience**. Here's a systematic approach to answer it confidently.

Step 1: Define "Slow"

Before jumping to solutions, ask clarifying questions:

- Is slow 500ms or 5 seconds? What's the acceptable threshold?
- Is it all endpoints or just one specific endpoint?
- Is it always slow or only under certain conditions (peak traffic, specific users, specific payload sizes)?

Step 2: Check Your Metrics

Open your observability tools — Datadog, Sentry, New Relic, etc. — and identify **which layer** is adding the latency. Is it the application layer, database, or an external service?

Step 3: Network Issues

Look for high latency, large payloads, or too many hops. Fixes:

- Use a **CDN** to serve static/cacheable content closer to users
- Enable **compression** (gzip/brotli) to reduce payload size
- Implement **pagination** to avoid large data dumps
- Apply **proper caching** (Redis, HTTP cache headers)

Step 4: Database Issues

The database is often the culprit. Check for:

- **Missing indexes** — add indexes on frequently queried columns
- **N+1 queries / queries in a loop** — batch them instead
- **Over-fetching** — select only the columns you need
- **Smart caching** — cache expensive query results

Step 5: External API Issues

If your API depends on third-party services:

- Replace **sequential calls** with **parallel calls** (Promise.all, async concurrency)
- Add **caching** for external responses where possible
- Set proper **timeouts** and implement **retry logic with backoff**
- Handle **rate limiting** gracefully

Step 6: Infrastructure Issues

If everything else looks fine, look at your infra:

- Low resources (CPU, memory) — **auto-scale**
- Connection pool limits — **tune your configuration**
- Check thread/worker limits on your server

Quick Summary Table

Layer	Problem	Fix
Network	High latency, large payload	CDN, compression, pagination
Database	Missing indexes, N+1 queries	Indexes, batch queries, caching
External API	Sequential calls, no timeout	Parallel calls, retry, caching
Infrastructure	Low resources, connection limits	Auto-scale, tune config